Example 3.4

Find the sum of the squares of the roots of $ax^4+bx^3+cx^2+dx+e = 0$, $a \neq 0$

Solution

Let $\alpha, \beta, \gamma,$ and δ be the roots of $ax^4+bx^3+cx^2+dx+e = 0$.

Then, we get

$$\sum_1 = \alpha+\beta+\gamma+\delta = -\frac{b}{a},$$

$$\sum_2 = \alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta = \frac{c}{a},$$

$$\sum_3 = \alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta = -\frac{d}{a},$$

$$\sum_4 = \alpha\beta\gamma\delta = \frac{e}{a}.$$

We have to find $\alpha^2+\beta^2+\gamma^2+\delta^2$.

Applying the algebraic identity

$$(a+b+c+d)^2 \equiv a^2+b^2+c^2+d^2+2(ab+ac+ad+bc+bd+cd),$$

we get

$$\alpha^2+\beta^2+\gamma^2+\delta^2 = (\alpha+\beta+\gamma+\delta)^2-2(\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta)$$

$$= \left(-\frac{b}{a}\right)^2-2\left(\frac{c}{a}\right)$$

$$= \frac{b^2-2ac}{a^2}.$$

Example 3.5

Find the condition that the roots of cubic equation $x^3+ax^2+bx+c = 0$ are in the ratio $p:q:r$.

Solution

Since roots are in the ratio $p:q:r$, we can assume the roots as $p\lambda, q\lambda$ and $r\lambda$.

Then, we get

$$\sum_1 = p\lambda+q\lambda+r\lambda = -a, \quad \ldots.(1)$$

$$\sum_2 = (p\lambda)(q\lambda)+(q\lambda)(r\lambda)+(r\lambda)(p\lambda) = b, \quad \ldots.(2)$$

$$\sum_3 = (p\lambda)(q\lambda)(r\lambda) = -c, \quad \ldots.(3)$$

Now, we get

$$(1) \Rightarrow \lambda = -\frac{a}{p+q+r} \quad \ldots.(4)$$

$$(3) \Rightarrow \lambda^3 = -\frac{c}{pqr} \quad \ldots..(5)$$

Substituting (4) in (5), we get

$$\left(-\frac{a}{p+q+r}\right)^3 = -\frac{c}{pqr} \Rightarrow pqra^3 = c(p+q+r)^3.$$

Example 3.6

Form the equation whose roots are the squares of the roots of the cubic equation $x^3+ax^2+bx+c = 0$.

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